Data set name	Brief description	Samples	Format	Usage
Omniglot	Different (around 1623) handwritten characters from 50 different alphabets	38300	Text, Images, Strokes	Classification, One-shot learning
Places365	There are 1.8 million training images from 365 scene categories in the Places365-Standard data set	Around 2 million	Images	Object classification, Scene recognition tasks
SBD (Semantic Boundaries Data set)	It currently contains annotations for each image, and provides both category-level and instance-level segmentations and boundaries. The segmentations and boundaries provided are for the 20 object categories in the VOC 2011 data set.	11355 images taken from the PASCAL VOC 2011 data set.	Images	Models for semantic contours prediction and semantic segmentation
STL10	This is an image recognition data set for developing unsupervised feature learning, deep learning, and self-taught learning algorithms.	100,000 unlabelled images for unsupervised learning, with 10 classes; images are 96x96 pixels in colour.	Images	Unsupervised learning
SVHN (Street View House Numbers)	This is a real-world image data set for developing machine learning and object recognition algorithms with minimal requirements of data preprocessing and formatting. SVHN is obtained from house numbers in Google Street View images.	Over 600,000 images	Images	Image classification
UCF101	This is an action recognition data set of realistic action videos, collected from YouTube, having 101 action categories. It is an extension of the UCF50 data set, which has 50 action categories.	13320 videos from 101 action categories	Videos	Action recognition
VOC (the PASCAL Visual Object Classes)	This data set consists of real images of complex scenes, including scale, pose, lighting and occlusion, for 20 classes — with complete annotation for all objects.	Around 10,000 images	Images with annotations	Object recognition and detection

Table 1: Large data sets

are movies. But a small proportion has also been obtained from open source databases such as the Prelinger archive, YouTube, and Google videos. It has around 7000 clips divided into 51 action categories, each containing over 101 clips, and can be used for object recognition and action detection.

- **Kinetics** is a collection of large-scale data sets of URL links of up to 650,000 video clips that cover various human action classes, depending on the data set version. The videos include human-object interactions and human-human interactions. Each action class has at least 400/600/700 video clips. Each clip is annotated, and can be used for object recognition and action detection.
- **LSUN** data set contains around one million labelled images for each of the 10 scene categories and 20 object categories. It can be used for understanding scenes with many ancillary tasks like room layout estimation, saliency prediction, etc.

Table 1 lists a few more large data sets that are available today.

Using the TensorFlow data sets (TFDS)

TFDS provides a beautiful collection of data sets that can be readily used in TensorFlow, Python and other machine learning frameworks.